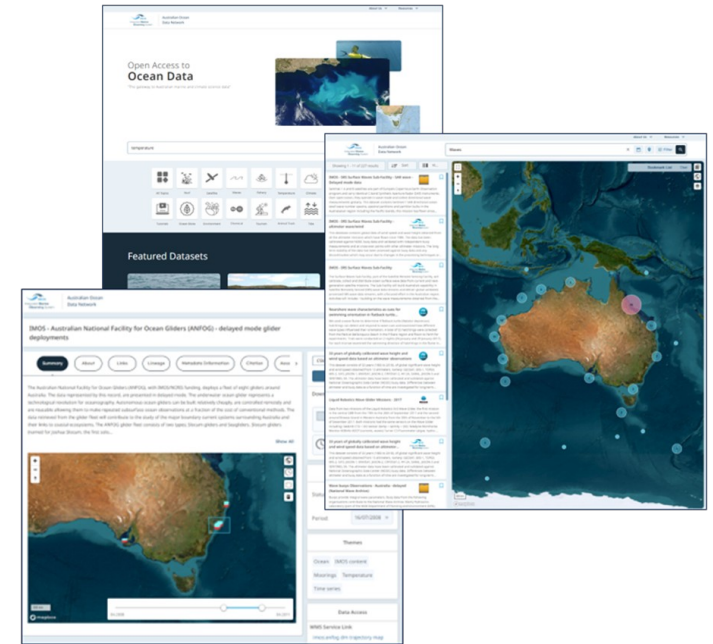


Preview of the New AODN Portal

July 2025

Vicky Isaac

Product Manager
Australian Ocean Data Network (AODN), Integrated Marine Observing System (IMOS)



IMOS acknowledges the Traditional Custodians and Elders of the land and sea on which we work and observe, and recognise them as Australia's first marine scientists and carers of Sea Country. We pay our respects to Aboriginal and Torres Strait Islander peoples past and present.

The new AODN Data Discovery Portal

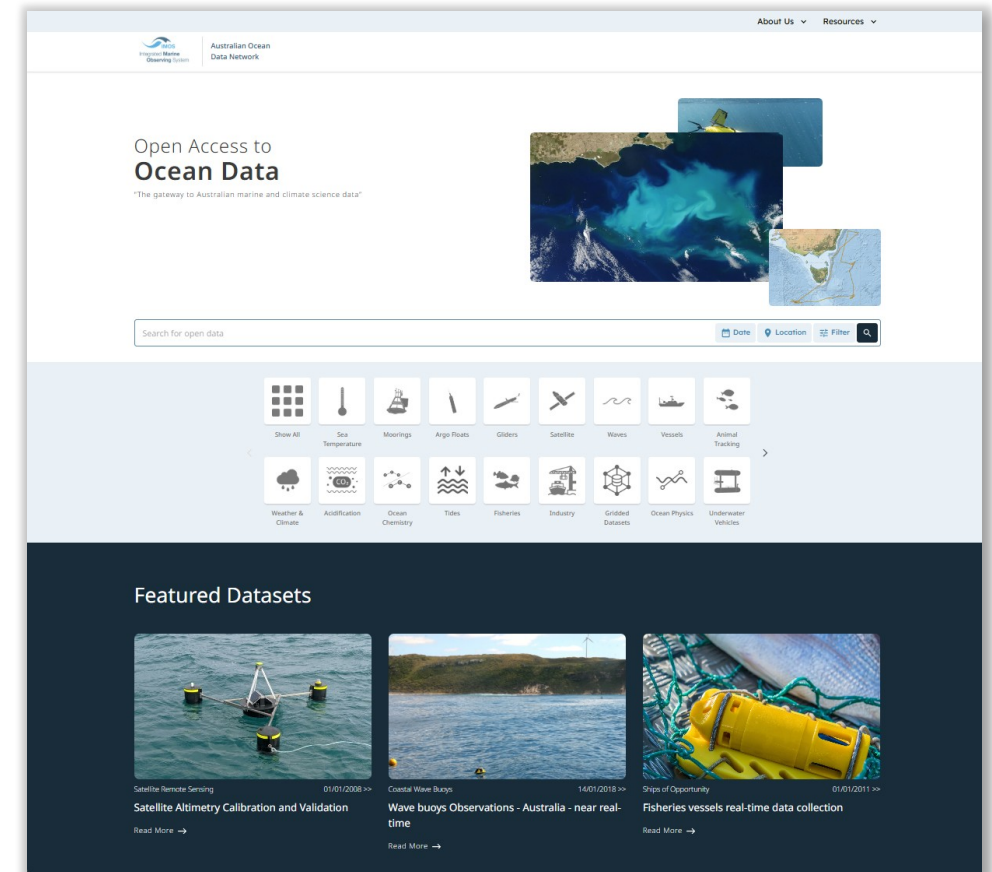
The purpose behind our portal rebuild

- We reached out to our user community to understand the key pain points of the existing portal, this is what they told us –
 - ⊗ Difficulty searching, filtering, finding the data that they want
 - ⊗ The usability of the website
 - ⊗ Only a small number of datasets available
 - ⊗ High threshold to make data available
 - ⊗ Had to use separate site to search the full national AODN catalogue
 - ⊗ In short - the current system, while functional, doesn't meet modern expectations.

The new AODN Data Discovery Portal

The purpose behind our portal rebuild

- ⦿ Increase the volume of discoverable data
- ⦿ Modernise the user experience
- ⦿ Make it easier for people to find data
- ⦿ Make it easier to access and use this data



We aim to deliver a single platform for the discoverability of Australia's Ocean Data assets

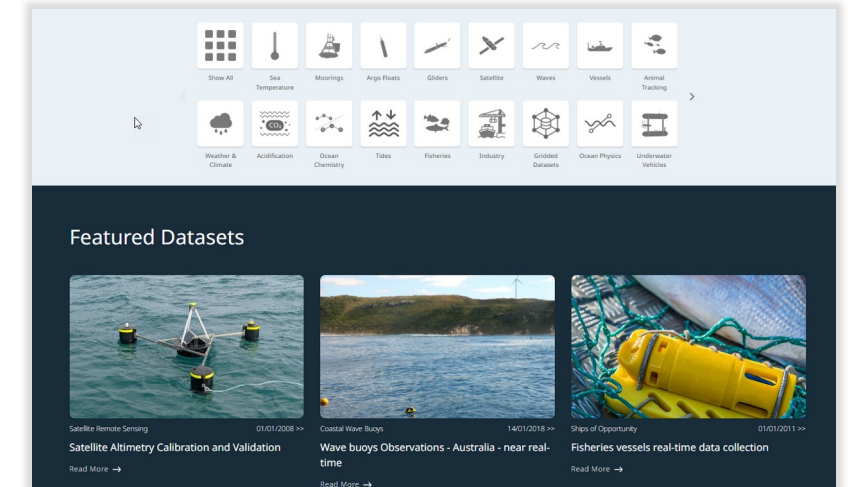
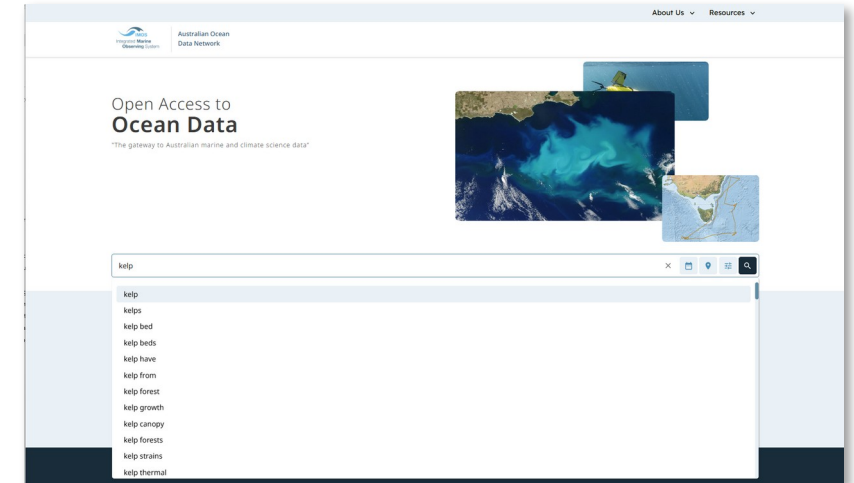
The new AODN Data Discovery Portal

Key Features

➤ Discoverability

- ★ Users can search all datasets described in the federated national catalogue ~ 14,000 datasets
- ★ Free text searching, google like search, search for anything, free text searching, fuzzy search
- ★ Topic searches, featured datasets

Helps you get started

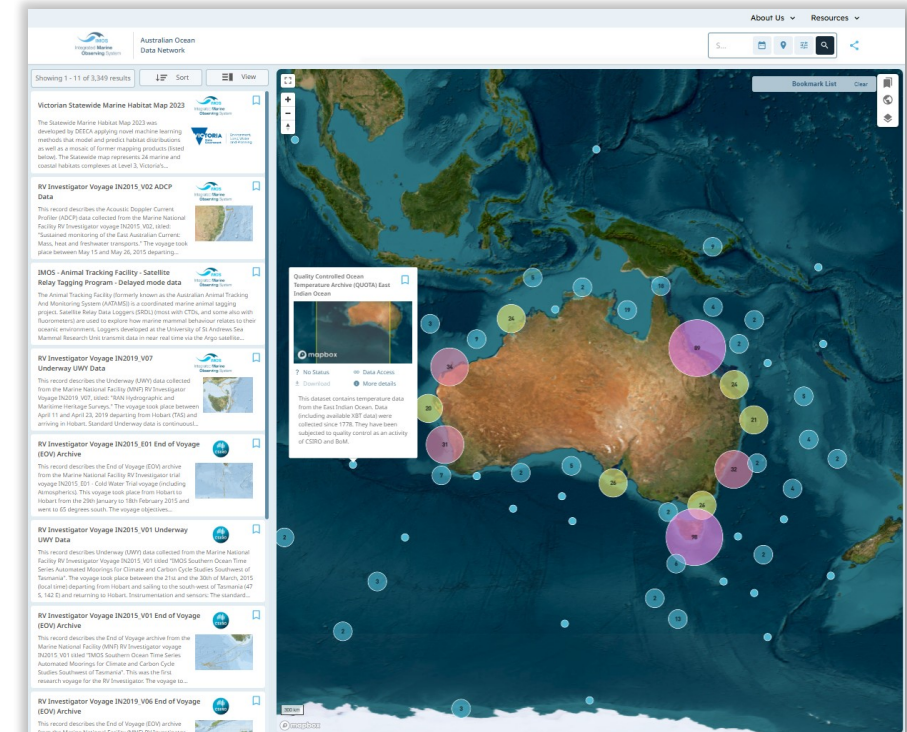
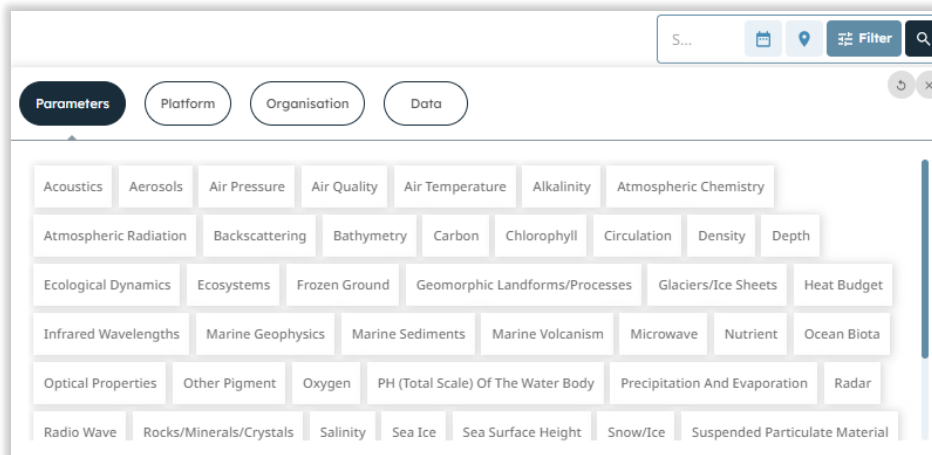


The new AODN Data Discovery Portal

Key Features

➤ Discoverability

- ★ Interactive maps to explore data -> search results update as you move around the map
- ★ Additional spatial, temporal and thematic filtering -> for more targeted searching

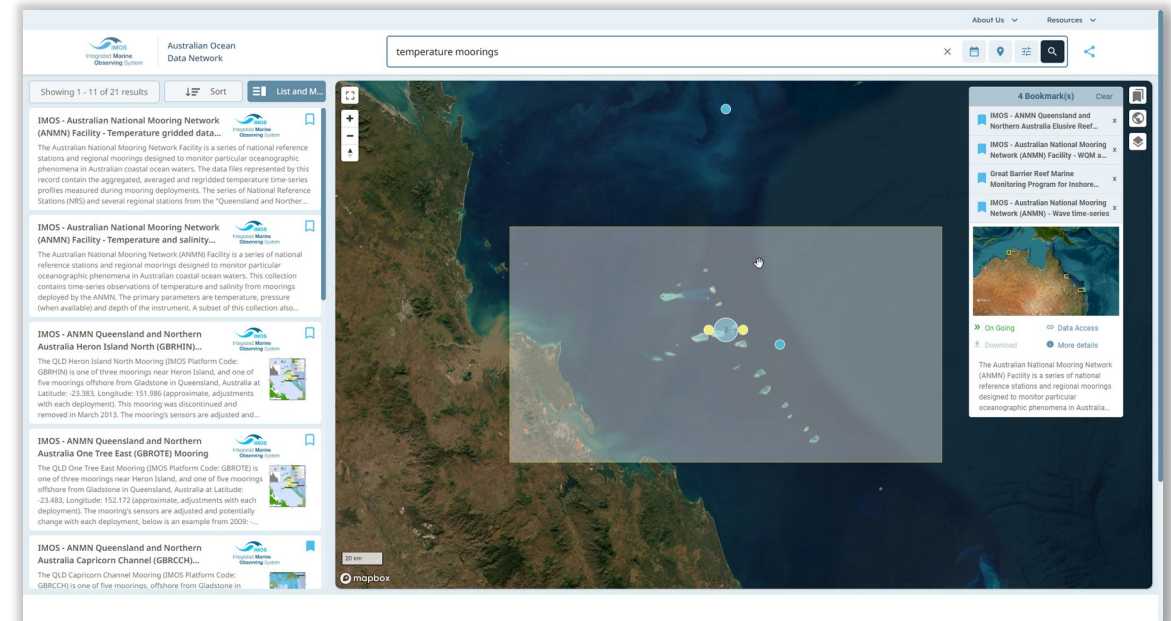
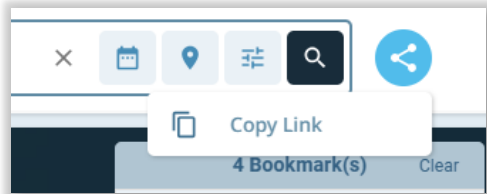


Helps you see what data is available and refine your search

The new AODN Data Discovery Portal

Key Features

- Personalisation and convenience
 - ★ Data preview
 - ★ Ability to bookmark multiple datasets
 - ★ Bookmarks saved across sessions
 - ★ Share and save search results



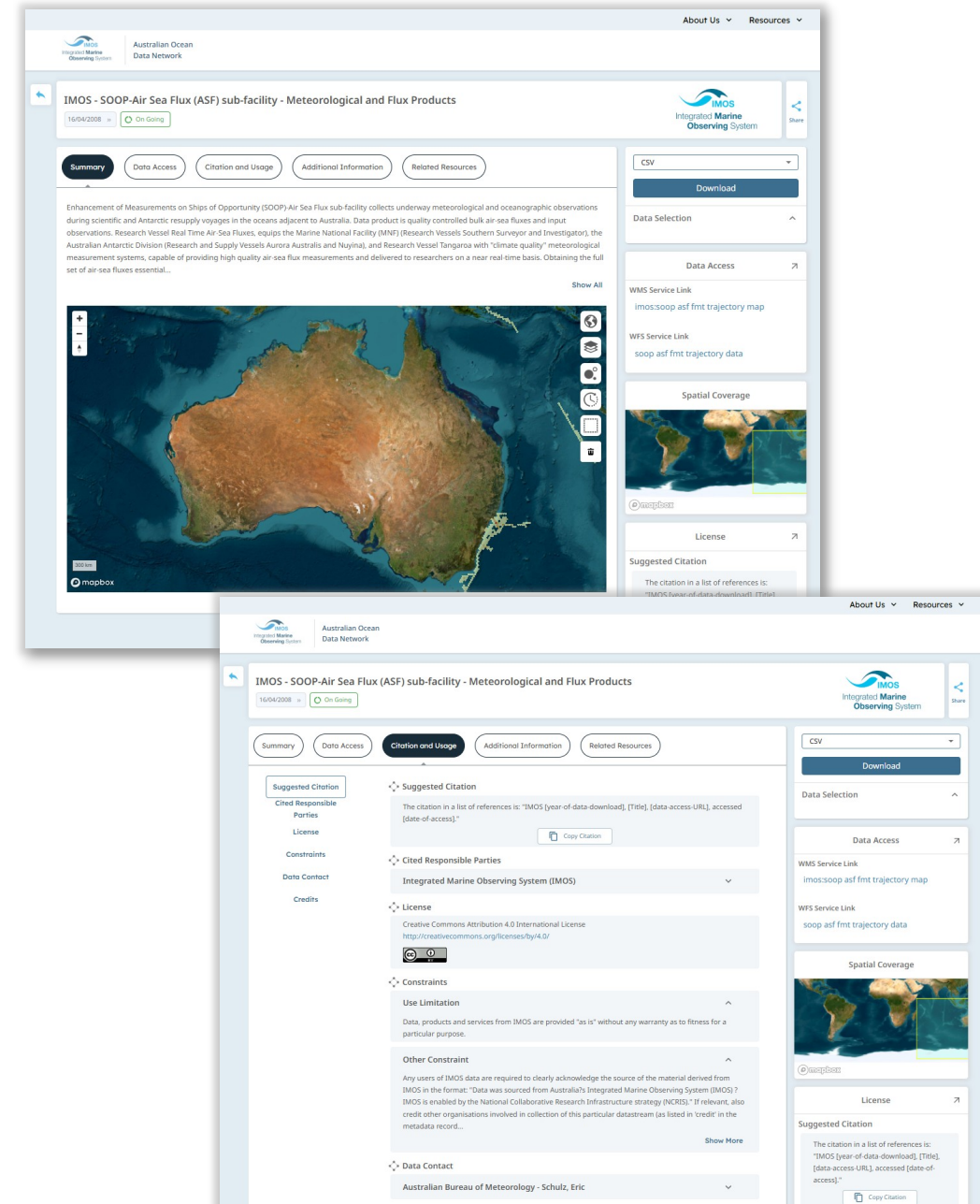
Helps save you time and collaborate with others

The new AODN Data Discovery Portal

Key Features

- Dataset Information
 - ☉ Clearly presented, easy to understand
 - ☉ What is it ? Who collected it? How was it collected ?
What was measured ? What does the data look like ?
 - ☉ Data license, citation and usage constraints
 - ☉ Dataset contact information
 - ☉ Data access methods and supporting documentation
 - ☉ Links to related data

Helps you decide - is this the data I am looking for ?

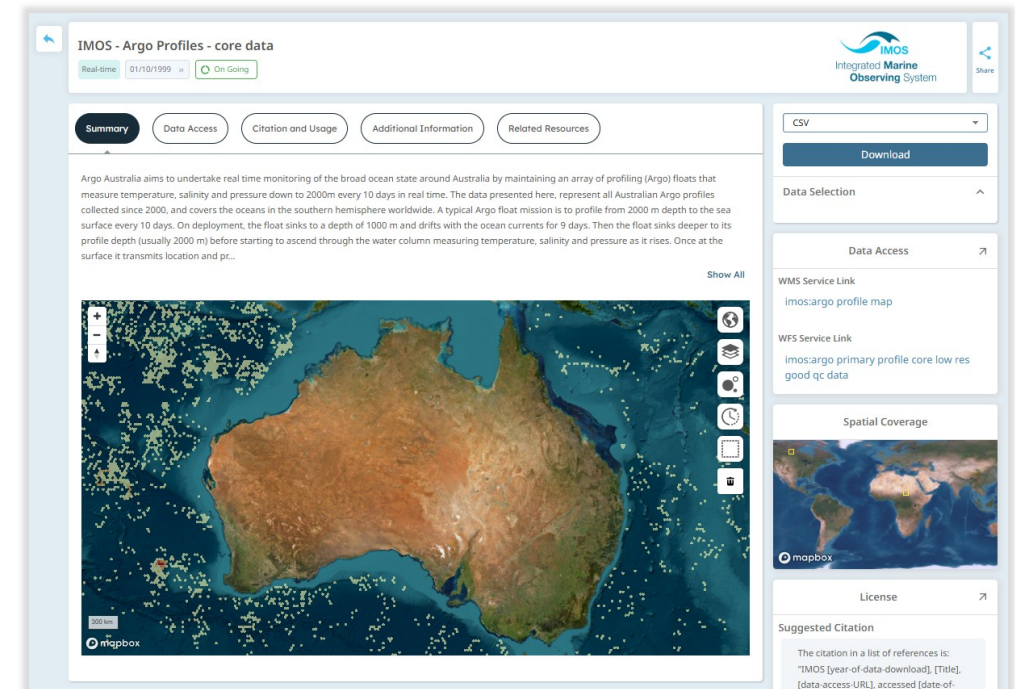
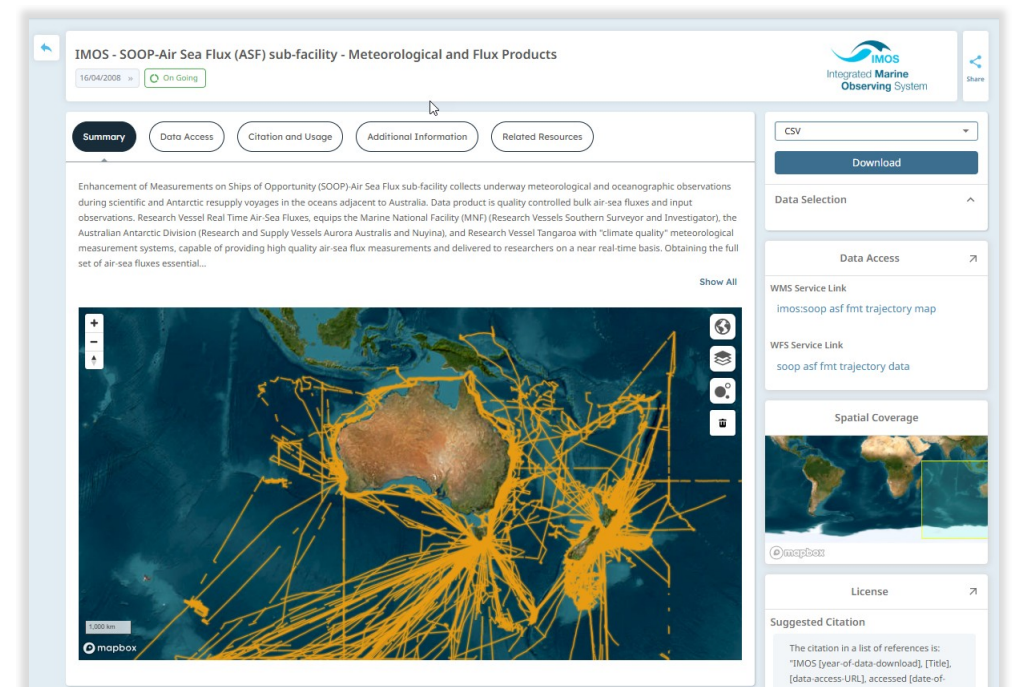


The new AODN Data Discovery Portal

Key Features

- Visualisations
 - 🕒 Geospatial visualisations of observation/measurement locations
 - 🕒 Data preview graphs - Timeseries / Point drill down – still to come

Helps you decide - is this the data I am looking for ?



The new AODN Data Discovery Portal

Key Features

➤ Data Access

- ❉ Various data access methods dependant on the dataset
- ❉ Interactive data filtering and download (aka subsetting) – data sent via email

The screenshot shows the AODN Data Discovery Portal interface. The top navigation bar includes 'About Us' and 'Resources'. The main content area displays the 'IMOS-SOOP Expendable Bathythermographs (XBT) Research Group - XBT real-time data' page. The page has tabs for 'Summary', 'Data Access', 'Citation and Usage', 'Additional Information', and 'Related Resources'. The 'Data Access' tab is active, showing a map of Australia with a bounding box (N: -32.45, S: -35.28, E: 155.33) and a date range (09-2022 to 01-2024). The map is labeled 'mapbox'. To the right of the map, there are links for 'WMS Service Link' and 'WFS Service Link'. Below the map, there is a 'Data Selection' section with a 'Bounding Box' and a 'Date Range' filter. The 'Data Access' section shows a 'WMS Service Link' and a 'WFS Service Link'. The 'Spatial Coverage' section shows a map of Australia with a bounding box.

	A	B		C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W
	Platform	site	TIME	LATITUDE	LONGITUDE	PL_DEPTH	PL_DEPTH	WIND	WIND	WIND	WIND	WIND	WIND	WIND	WIND	WIND	WIND	WIND	WIND	WIND	WIND	WIND	WIND	
1	IMOS	SOOP	INVESTIGATOR	4700.0	39.2726	148.917	180	43.3	4.7	254.7	11.3	24.8	1	1056.7	20.6	1	14.4	24.4	2	75.6	24.4	2	16.2560	5.9
2	IMOS	SOOP	INVESTIGATOR	4700.0	39.2726	148.917	180	43.3	4.7	254.7	11.3	24.8	1	1056.7	20.6	1	14.4	24.4	2	75.6	24.4	2	16.2560	5.9
3	IMOS	SOOP	INVESTIGATOR	4700.0	39.2726	148.917	180	43.3	4.7	254.7	11.3	24.8	1	1056.7	20.6	1	14.4	24.4	2	75.6	24.4	2	16.2560	5.9
4	IMOS	SOOP	INVESTIGATOR	4700.0	39.2726	148.917	180	43.3	4.7	254.7	11.3	24.8	1	1056.7	20.6	1	14.4	24.4	2	75.6	24.4	2	16.2560	5.9
5	IMOS	SOOP	INVESTIGATOR	4700.0	39.2726	148.917	180	43.3	4.7	254.7	11.3	24.8	1	1056.7	20.6	1	14.4	24.4	2	75.6	24.4	2	16.2560	5.9
6	IMOS	SOOP	INVESTIGATOR	4700.0	39.2726	148.917	180	43.3	4.7	254.7	11.3	24.8	1	1056.7	20.6	1	14.4	24.4	2	75.6	24.4	2	16.2560	5.9
7	IMOS	SOOP	INVESTIGATOR	4700.0	39.2726	148.917	180	43.3	4.7	254.7	11.3	24.8	1	1056.7	20.6	1	14.4	24.4	2	75.6	24.4	2	16.2560	5.9
8	IMOS	SOOP	INVESTIGATOR	4700.0	39.2726	148.917	180	43.3	4.7	254.7	11.3	24.8	1	1056.7	20.6	1	14.4	24.4	2	75.6	24.4	2	16.2560	5.9
9	IMOS	SOOP	INVESTIGATOR	4700.0	39.2726	148.917	180	43.3	4.7	254.7	11.3	24.8	1	1056.7	20.6	1	14.4	24.4	2	75.6	24.4	2	16.2560	5.9
10	IMOS	SOOP	INVESTIGATOR	4700.0	39.2726	148.917	180	43.3	4.7	254.7	11.3	24.8	1	1056.7	20.6	1	14.4	24.4	2	75.6	24.4	2	16.2560	5.9
11	IMOS	SOOP	INVESTIGATOR	4700.0	39.2726	148.917	180	43.3	4.7	254.7	11.3	24.8	1	1056.7	20.6	1	14.4	24.4	2	75.6	24.4	2	16.2560	5.9
12	IMOS	SOOP	INVESTIGATOR	4700.0	39.2726	148.917	180	43.3	4.7	254.7	11.3	24.8	1	1056.7	20.6	1	14.4	24.4	2	75.6	24.4	2	16.2560	5.9
13	IMOS	SOOP	INVESTIGATOR	4700.0	39.2726	148.917	180	43.3	4.7	254.7	11.3	24.8	1	1056.7	20.6	1	14.4	24.4	2	75.6	24.4	2	16.2560	5.9
14	IMOS	SOOP	INVESTIGATOR	4700.0	39.2726	148.917	180	43.3	4.7	254.7	11.3	24.8	1	1056.7	20.6	1	14.4	24.4	2	75.6	24.4	2	16.2560	5.9
15	IMOS	SOOP	INVESTIGATOR	4700.0	39.2726	148.917	180	43.3	4.7	254.7	11.3	24.8	1	1056.7	20.6	1	14.4	24.4	2	75.6	24.4	2	16.2560	5.9
16	IMOS	SOOP	INVESTIGATOR	4700.0	39.2726	148.917	180	43.3	4.7	254.7	11.3	24.8	1	1056.7	20.6	1	14.4	24.4	2	75.6	24.4	2	16.2560	5.9
17	IMOS	SOOP	INVESTIGATOR	5200.0	39.2818	148.906	180	43.2	4.7	257.1	10.8	24.8	1	1056.7	20.6	1	14.5	24.4	2	75.2	24.4	2	16.3444	5.9
18	IMOS	SOOP	INVESTIGATOR	5200.0	39.2818	148.906	180	43.2	4.7	257.1	10.8	24.8	1	1056.7	20.6	1	14.5	24.4	2	75.2	24.4	2	16.3444	5.9
19	IMOS	SOOP	INVESTIGATOR	5200.0	39.2818	148.906	180	43.2	4.7	257.1	10.8	24.8	1	1056.7	20.6	1	14.5	24.4	2	75.2	24.4	2	16.3444	5.9
20	IMOS	SOOP	INVESTIGATOR	5200.0	39.2818	148.906	180	43.2	4.7	257.1	10.8	24.8	1	1056.7	20.6	1	14.5	24.4	2	75.2	24.4	2	16.3444	5.9
21	IMOS	SOOP	INVESTIGATOR	5200.0	39.2818	148.906	180	43.2	4.7	257.1	10.8	24.8	1	1056.7	20.6	1	14.5	24.4	2	75.2	24.4	2	16.3444	5.9
22	IMOS	SOOP	INVESTIGATOR	5200.0	39.2818	148.906	180	43.2	4.7	257.1	10.8	24.8	1	1056.7	20.6	1	14.5	24.4	2	75.2	24.4	2	16.3444	5.9
23	IMOS	SOOP	INVESTIGATOR	5200.0	39.2818	148.906	180	43.2	4.7	257.1	10.8	24.8	1	1056.7	20.6	1	14.5	24.4	2	75.2	24.4	2	16.3444	5.9

finish processing data file whose uuid is: 07818819-2e56-4a12-9395-0082b57b2fe8

The result is ready.

The id of the dataset is: 07818819-2e56-4a12-9395-0082b57b2fe8

The condition(s): start date: 09-2017, end date: 02-2025, polygon: ("type": "MultiPolygon", "coordinates": [[[[[146.8769885110496, -40.73340867863572], [149.38153156123204, -40.73340867863572], [149.38153156123204, -39.25942279433798], [146.8769885110496, -39.25942279433798], [146.8769885110496, -40.73340867863572]]]]], .

You can download it. The download link is: <https://havier-example-bucket.s3.ap-southeast-2.amazonaws.com/07818819-2e56-4a12-9395-0082b57b2fe8/date:09-2017-02-2025.csv>

Dataset Download

1 Input your email address 2 Usage limitation and licenses

Data Selection

Bounding Box

N: -32.45 W: 151.24

S: -35.28 E: 155.33

Date Range

09-2022 to 01-2024

Your Email Address

Email Address *

Processing dataset download may take some time. It is varied by the size of the dataset, the selected conditions, and the server load (may take several seconds to several hours or even more). Please provide your email address to receive the download link and necessary information.

Data Usage Information (optional)

a. For what purpose do you intend to use the data?

☐ Education ☐ Scientific research ☐ Industry operations ☐ Recreational

b. If not recreational - what sector do you work for?

☐ Academic ☐ Industry ☐ Government ☐ Other

c. Do you consent to being contacted by IMOS to further understand your data needs and the use of our products and services?

☐ Yes ☐ No

Next

The new AODN Data Discovery Portal

Key Features

- Data Access
 - Links to data services e.g. THREDDS, WMS, WFS
 - Links to other data sources (non-AODN hosted datasets)
 - Advanced data access support e.g. Python notebooks and tutorials

IMOS - SOOP-Air Sea Flux (ASF) sub-facility - Meteorological and Flux Products

16/04/2028 On Going

Summary Data Access Citation and Usage Additional Information Related Resources

Data Access Options

Links

- Air sea flux page on IMOS website
- Quality control procedure for IMOS real-time meteorological and sea surface observations, and air-sea fluxes from research vessel and mooring platforms - r
- IMOS ship SST for satellite SST validation - report (RV Investigator)
- NetCDF files via THREDDS catalog
- View and download data through the AODN Portal

Show More +

Data Access List

- imos:soop asf fnt trajectory map
- soop asf fnt trajectory data

Python Notebook

- Access to jupyter notebook to query Cloud Optimised converted dataset
- Python notebook example

Jupyter notebook interface:

Access Vessel Air Sea Flux Product Delayed (Parquet)

This Jupyter notebook demonstrates how to access and plot vessel air sea flux product delayed data, available as a Parquet dataset stored on S3.

More information about the dataset is available in the AODN metadata catalogue.

The source of truth for this notebook is maintained on GitHub.

```
In [1]: dataset_name = "vessel_air_sea_flux_product_delayed"
```

Install/Update packages and Load common functions

```
In [2]: import os, requests, importlib.util

open('setup.py', 'w').write(requests.get("https://raw.githubusercontent.com/aodn/aodn_cloud_optimised/main/notebooks/setup.py").text)

spec = importlib.util.spec_from_file_location("setup", "setup.py")
setup = importlib.util.module_from_spec(spec)
spec.loader.exec_module(setup)
setup.install_requirements()
setup.load_dataquery()
```

Using Python 3.12.6 Interpreter at: /home/ibessard/miniforge3/envs/aodnCloudOptimised/bin/python

Creating virtual environment at: .env

Activate with: source .env/bin/activate

Audited 246 packages in 53ms

Local version 0.2.1 is up to date (remote: 0.2.1)

```
In [3]: from DataQuery import getdata
```

Australian Ocean Data Network

16/04/2028 On Going

Summary Data Access Citation and Usage Additional Information Related Resources

Data Access Options

Links

- DATA ACCESS - CoTS movement summaries (CSV direct download)
- DATA ACCESS - CoTS individual movement tracks (per image) from time-lapse photography (CSV direct download)
- DATA ACCESS - feeding electivity on coral species by CoTS from time-lapse photography (CSV direct download)

Spatial Coverage

License

Suggested Citation

Suggested Citation Not Found

Australian National Fish Collection (ANFC)

Latest version published by CSIRO National Collections and Marine Infrastructure (NCMI) Information and Data Centre (IDC) on Nov 30, 2023

Download the latest version of this resource data as a Darwin Core Archive (DwC-A) or the resource metadata as EML or RTT.

Data as a DwC-A file

Metadata as an EML file

Metadata as an RTT file

Home / Resource

Occurrence Specimen

Download 46,130 records in English (3 MB) - Update frequency: Irregular

Download in English (10 KB)

Download in English (9 KB)

Home: <http://www.marlin.csiro.au/gisnetwork/en/>

GBIF UUID: [5f7402543a9-9f7b525755a](https://doi.org/10.25919/5f7402543a9-9f7b525755a)

Publication date: 30 November 2023

Published by: CSIRO National Collections and Marine Infrastructure (NCMI) Information and Data Centre

IDC

License: CC BY 4.0

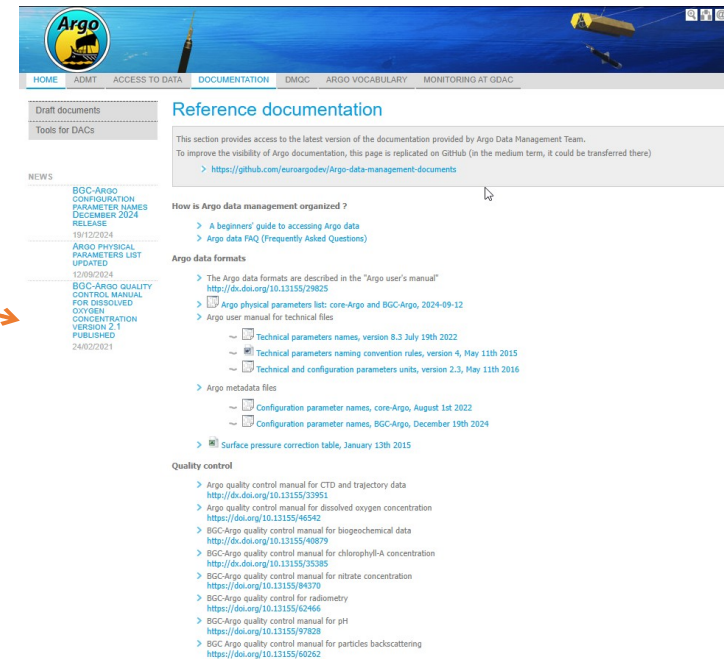
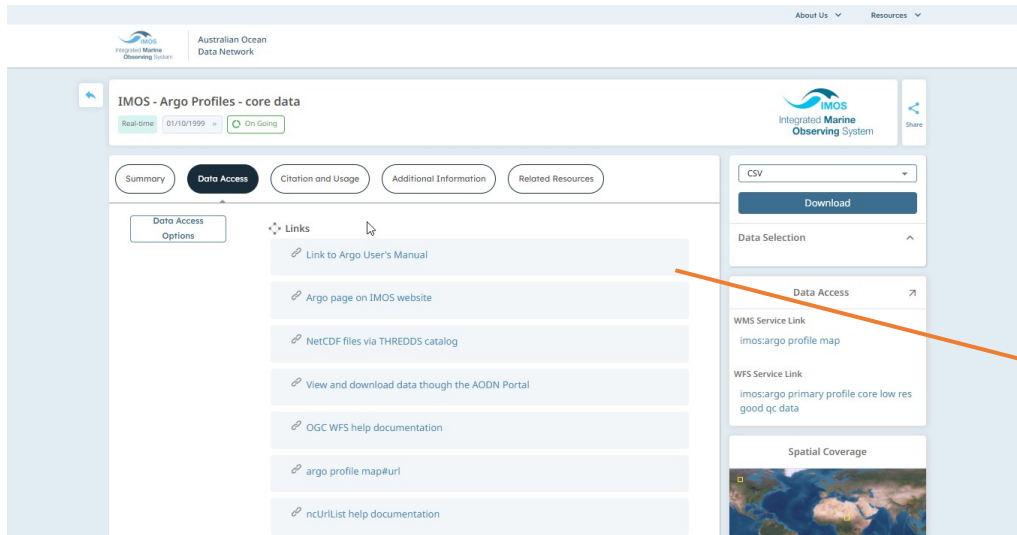
View as file

Location	Site	Latitude	Longitude	Date	start	Depth	Starfish	s	FDV	area	Day	Fract	Day	Fract	Monitored	Displace	Displace	Time	Mean	Sim	Max	Simel	Proportion	Time	Need	Emergence	Homing	L	Live
2	Cairns	Outf	-14.68	145.4436	#####	3.5	370	1.166257	0.522778	0.538333	1960	15	18.62	9	180	25.01	136.96	0.78	140	1	0	0	0	0	0	0	0	0	
3	Cairns	Outf	-14.6796	145.4440	#####	2.5	200	1.6420	0.480806	0.447917	1384	5.5	5.72	33	660	9.05	191.88	0	0	1	0	0	0	0	0	0	0	0	
4	Cairns	Outf	-14.6799	145.4440	#####	4	290	0.879607	0.4375	0.361104	1333	1.5	1.62	34	680	13.45	152.34	0.5	340	1	0	0	0	0	0	0	0	0	
5	Cairns	Outf	-14.68	145.4444	#####	3	250	0.557283	0.45625	0.371328	1318	0.55	0.6	41	820	2.42	43.92	0	0	0	0	0	0	0	0	0	0	0	
6	Cairns	Outf	-14.6803	145.4448	#####	2.5	300	0.937403	0.452778	0.481309	1306	0.2	0.21	47	940	4.79	52.2	0.38	360	0	0	0	0	0	0	0	0	0	
7	Cairns	Outf	-14.6803	145.4479	#####	2.5	300	0.561189	0.703333	0.475684	1200	4.1	4.52	1	20														
8	Cairns	Outf	-14.677	145.4448	#####	2.5	270	0.857071	0.526389	0.539722	1200	1.7	2.04	12	240	13.84	84	0	0	0	0	0	0	0	0	0	0	0	
9	Cairns	Outf	-14.6803	145.4484	#####	2.5	300	0.894440	0.645833	0.449667	960	0.2	0.29	27	540	8.51	94.2	0.37	200	0	0	0	0	0	0	0	0	0	
10	Cairns	Outf	-14.6873	145.4671	#####	2.5	300	0.826875	0.708333	0.475917	1101	0.2	0.26	55	1100	4.83	52.86	0.35	380	0	0	0	0	0	0	0	0	0	
11	Cairns	Pathway	-14.6977	145.4473	#####	2	200	0.869947	0.589583	0.381344	1141	13.7	17.39	49	980	5.08	104.15	0	0	1	0	0	0	0	0	0	0	0	
12	Cairns	Pathway	-14.6966	145.4473	#####	6	320	0.801557	0.395833	0.259028	1487	0.8	0.77	17	340	11	86.9	0	0	1	0	0	0	0	0	0	0	0	
13	Cairns	Pathway	-14.6967	145.4473	#####	3.5	300	0.509232	0.378472	0.355556	1407	5.1	5.22	45	900	5.28	106.14	0	0	1	0	0	0	0	0	0	0	0	
14	Cairns	Pathway	-14.6968	145.4471	#####	4	250	0.657954	0.375	0.354367	1410	0.2	0.2	60	1200	2.14	61.8	0.37	440	0	1	0	0	0	0	0	0	0	
15	Cairns	Pathway	-14.6968	145.4473	#####	4.2	330	1.518714	0.402778	0.365972	1487	0.3	0.29	60	1200	7.86	89.64	0.3	360	0	1	0	0	0	0	0	0	0	
16	Cairns	Pathway	-14.6966	145.4479	#####	4.2	200	0.794131	0.644444	0.395833	1082	4.6	6.12	32	640	5.11	85.5	0.44	280	1	0	0	0	0	0	0	0	0	
17	Cairns	Pathway	-14.6966	145.4481	#####	3.5	230	1.406648	0.696258	0.402778	937	0.2	0.31	55	1100	2.95	68.15	0.58	640	0	0	0	0	0	0	0	0	0	
18	Cairns	Pathway	-14.6968	145.4473	#####	1.8	350	0.561189	0.392961	0.38125	1484	0.2	0.21	60	1200	1.7	29.15	0.31	440	0	1	0	0	0	0	0	0	0	
19	Cairns	Pathway	-14.6966	145.4473	#####	2	110	1.362828	0.391667	0.380417	1471	0.25	0.25	60	1200	5.95	71.64	0.35	420	0	1	0	0	0	0	0	0	0	
20	Cairns	Pathway	-14.6959	145.4492	9/06/2015	2	290	1.052103	0.625	0.449306	1187	0.2	0.24	59	1180	3.72	53.56	0.42	380	0	1	0	0	0	0	0	0	0	
21	Cairns	Pathway	-14.6959	145.4492	9/06/2015	3	310	0.817113	0.612578	0.447222	1190	0.1	0.12	60	1200	5.36	85.14	0.43	520	0	1	0	0	0	0	0	0	0	
22	Cairns	Pathway	-14.6959	145.4492	9/06/2015	3.8	210	0.683398	0.620139	0.448611	1193	0.2	0.24	60	1200	3.04	88.44												
23	Cairns	Pathway	-14.6961	145.449	9/06/2015	2.5	180	1.104134	0.448611	0.364083	1199	0.1	0.12	36	320	5.47	53.56	0.88	280	0	1	0	0	0	0	0	0	0	
24	Cairns	Pathway	-14.6961	145.4489	9/06/2015	2.5	225	1.123408	0.465278	0.398028	1190	0.65	0.81	45	900	5.68	134.1	0	0	1	0	0	0	0	0	0	0	0	
25	Cairns	Pathway	-14.696	145.4489	9/06/2015	4	550	1.241012	0.631944	0.381111	1020	0.1	0.14	59	1180	9.92	217.38	0.47	560	0	1	0	0	0	0	0	0	0	
26	Cairns	Pathway	-14.6961	145.448	9/06/2015	3.8	255	0.442963	0.447222	0.366667	1204	0.1	0.12	59	1180	2.8	25.5												
27	Cairns	Pathway	-14.6959	145.4482	9/06/2015	2.5	295	0.727127	0.375	0.3	1200	0.2	0.24	0	0														

The new AODN Data Discovery Portal

Key Features

- Supporting Information about the Data
 - 🕒 Contact information
 - 🕒 Supporting documentation

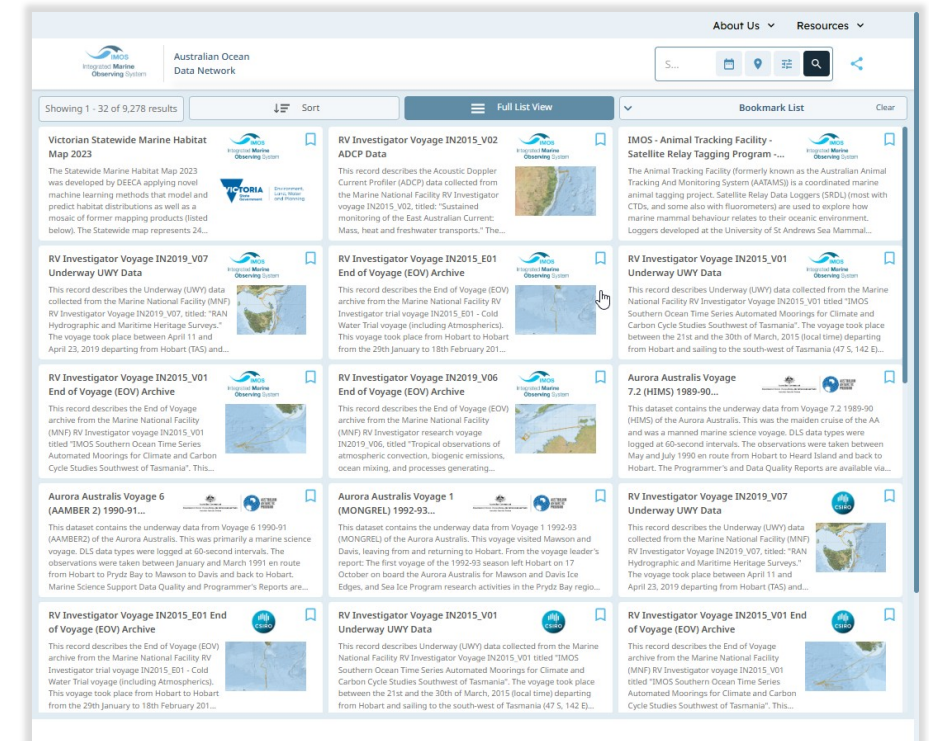
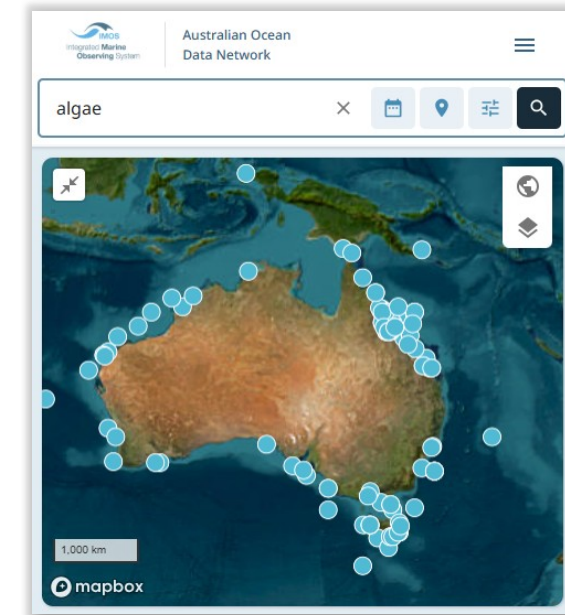
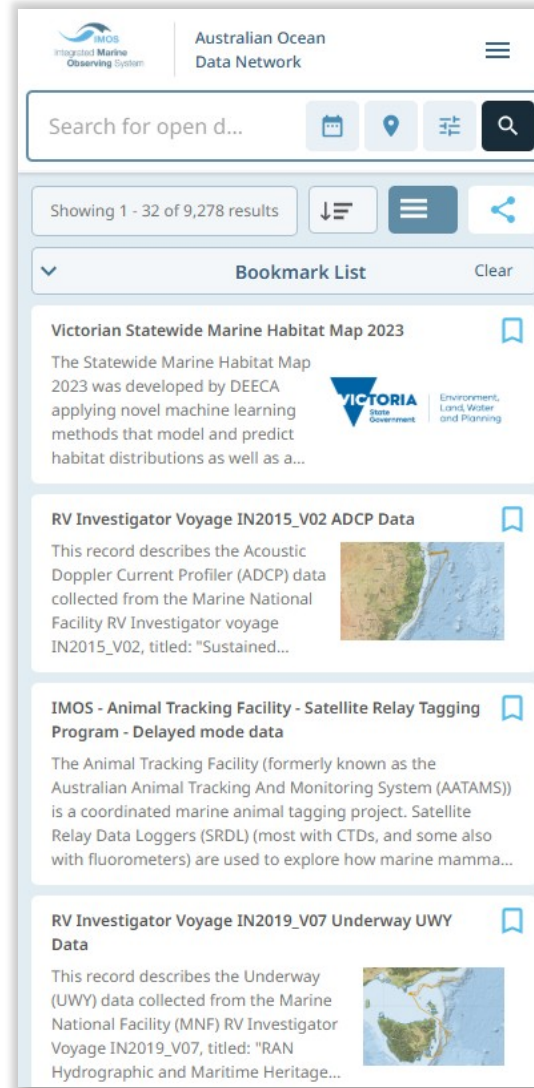


The new AODN Data Discovery Portal

Key Features

➤ Usability

- Modern look and feel
- Improved navigation
- Adapts across devices
- Improved accessibility



Questions?

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